

S = Silhouette coefficient

$S = 1 - a/b$ if $a < b$

Typically between 0 & 1

closer to 1 $\Rightarrow$ better

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Performance Metrics for Clustering -

Silhouette Score:

- 1) Measures how ~~well~~ similar a data point is to its own cluster compared to other clusters

$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

$$S = 1 - a/b \text{ if } a < b$$

where, $a(i)$ = avg distance between point i and other points in the same cluster

$b(i)$ = min average distance between points i and points in other clusters

- 2) Interpretation:

$S(i) \approx 1$: Well clustered

$S(i) \approx 0$: On or near cluster boundary

$S(i) < 0$: Misclassified

eg- Point i in cluster with $a(i) = 2$, nearest other cluster $b(i) = 5$

$$S(i) = \frac{5 - 2}{\max(2, 5)} = \frac{3}{5} = 0.6$$

Moderately well clustered

Association Rule Learning -

- 1) Unsupervised learning technique used to discover interesting relationships, patterns, or associations between items in large datasets

- 2) Support:

- Measures how frequently the itemset appears in the dataset

$$\text{Support}(X) = \frac{\text{No. of transactions containing } X}{\text{Total no. of transactions}}$$

3) Confidence:

- Confidence of a rule $X \Rightarrow Y$ measures how often items in Y appear in transactions that contain X

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

4) Lift:

- Measures strength of a rule over random chance.

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Confidence}(X \Rightarrow Y)}{\text{Support}(Y)}$$

Interpretation:

Lift > 1 : Positive Correlation between X and Y

Lift $= 1$: No correlation

Lift < 1 : Negative Correlation

Apriori Algorithm -

1) Iterative algorithm to find frequent itemsets

2) Steps:

(i) Identify all frequent 1 itemsets with support $\geq$ min support ~~has~~ threshold

(ii) Generate candidate k itemsets from frequent $(k-1)$ itemsets

(iii) Prune candidates containing infrequent subsets

(iv) Repeat until no k new frequent itemsets are found

(v) Generate rules from frequent itemsets that satisfy minimum confidence

Limitations -

1) Multiple Database Scans

2) Vast Candidate Generation

3) High Computational Cost

Eg. Transactions:

$T_1: \{A, B, C\}$, $T_2: \{A, B\}$, $T_3: \{A, C\}$, $T_4: \{B, C\}$,
 $T_5: \{A, B, C\}$

(i) Support 1 itemsets:

$A: 4/5 = 0.8$, $B: 4/5 = 0.8$, $C: 4/5 = 0.8$

(ii) Generate 2 itemsets:

$\{A, B\}: 3/5 = 0.6$, $\{A, C\}: 3/5$, $C: 4/5 = 0.8$

(iii) Generate 3 itemsets:

$\{A, B, C\}: 2/5 = 0.4$

Assuming min support = 0.5 $\Rightarrow$ 3 itemset discarded

(iv) Generate rules:

$A \Rightarrow B: \text{Confidence} = \frac{\text{Support}(A, B)}{\text{Support}(A)} = \frac{0.6}{0.8} = 0.75$

F-P Growth Algorithm (Frequent Pattern Growth) —

1) ~~Alternate~~ Alternative to Apriori; avoids candidate generation

2) Steps:

(i) Construct an FP tree from the dataset, storing item frequencies & paths

(ii) Recursively extract frequent patterns by traversing FP tree

(iii) Generate association rules from frequent patterns using confidence thresholds

3) Faster than Apriori for large datasets with long frequent patterns

Advantages —

1) Avoids candidate generation

2) Requires only 2 database scans

3) Reduces I/O cost

4) More memory-efficient than Apriori

5) Uses compact FP tree

6) Scales well with big data

7) Faster on large & dense datasets

Divide & Conquer

Applications of Association Rule Learning-

- 1) Market Basket Analysis: Discovering which products are often bought together
- 2) Cross selling strategies: Suggesting complimentary products
- 3) Medical diagnosis: Finding associations between symptoms & diseases
- 4) Web usage mining: Understanding browsing patterns on websites

Q. Using the Apriori Algorithm, find all frequent 2 itemsets for the transactions:

$T_1: \{A, B\}$, $T_2: \{A, C\}$, $T_3: \{B, C\}$, $T_4: \{A, B\}$, $T_5: \{B, C\}$,
Min Support = 0.4

Ans Step 1: Find support of 1 itemsets:

A: $T_1, T_2, T_4 = 3/5 = 0.6 \geq 0.4$ frequent

B: $T_1, T_3, T_4, T_5 = 4/5 = 0.8 \geq 0.4$ frequent

C: $T_2, T_3, T_5 = 3/5 = 0.6 \geq 0.4$ frequent

Step 2: Generate candidate 2 itemsets:

$\{A, B\}$, $\{A, C\}$, $\{B, C\}$

Step 3: Compute support:

$\{A, B\}: T_1, T_4 = 2/5 = 0.4$ frequent

$\{A, C\}: T_2 = 1/5 = 0.2$ not frequent

$\{B, C\}: T_3, T_5 = 2/5 = 0.4$ frequent

Frequent 2 itemsets:

$\{A, B\}$, $\{B, C\}$

Q. Calculate the Confidence for the rule $A \Rightarrow B$ using the above transactions

Ans
$$\text{Confidence}(A \Rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} = \frac{2/5}{3/5} = \frac{2}{3} \approx 0.667$$

$\Rightarrow$ When A occurs in a transaction, there is a 66.7% chance that B also occurs

Challenges & Limitations of Association Rule Learning -

- 1) Generates a very large no. of rules, many of which may be irrelevant
- 2) Difficult to set appropriate support & confidence thresholds
- 3) May miss rare but important item associations
- 4) Computationally expensive for large & high dimensional datasets
- 5) Does not capture sequential or temporal relationships between items
- 6) Rules may lack practical significance
- 7) Hard to interpret
- 8) Sensitive to noisy & incomplete data

Market Basket Analysis

- 1) Identify purchase patterns
- 2) Categorize customer purchase behavior
- 3) Identify actionable information
- 4) Benefits:
 - (i) Selection of promotion, merchandising strategy
 - (ii) Uncover consumer spending patterns
 - (iii) Joint promotional opportunities
 - (iv) Single computations
 - (v) Can be undirected
 - (vi) Different data forms can be analyzed

Basket data = transaction data

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5) Applications:

- (i) Retail outlets
- (ii) Telecommunications
- (iii) Banks
- (iv) Insurance
- (v) Medical

6) Limitations:

- (i) Takes over 18 months to implement
- (ii) Market basket analysis only identifies hypotheses, which need to be tested
- (iii) Measurement of impact needed
- (iv) Difficult to identify product groupings
- (v) Complexity grows exponentially

Reinforcement Learning -

- 1) A learning paradigm where an agent learns to make decisions by interacting with an environment to maximize cumulative reward.
- 2) Does not rely on labeled input/output pairs.
- 3) Agent learns through trial and error: actions that lead to higher rewards are reinforced, while actions ~~are~~ with lower rewards are discouraged.
- 4) Types:
 - (i) Model Based RL
 - (ii) Model Free RL

Model Based RL -

- 1) The agent builds a model of the environment
- 2) Uses the model to plan future actions
- 3) eg- Dyna-Q

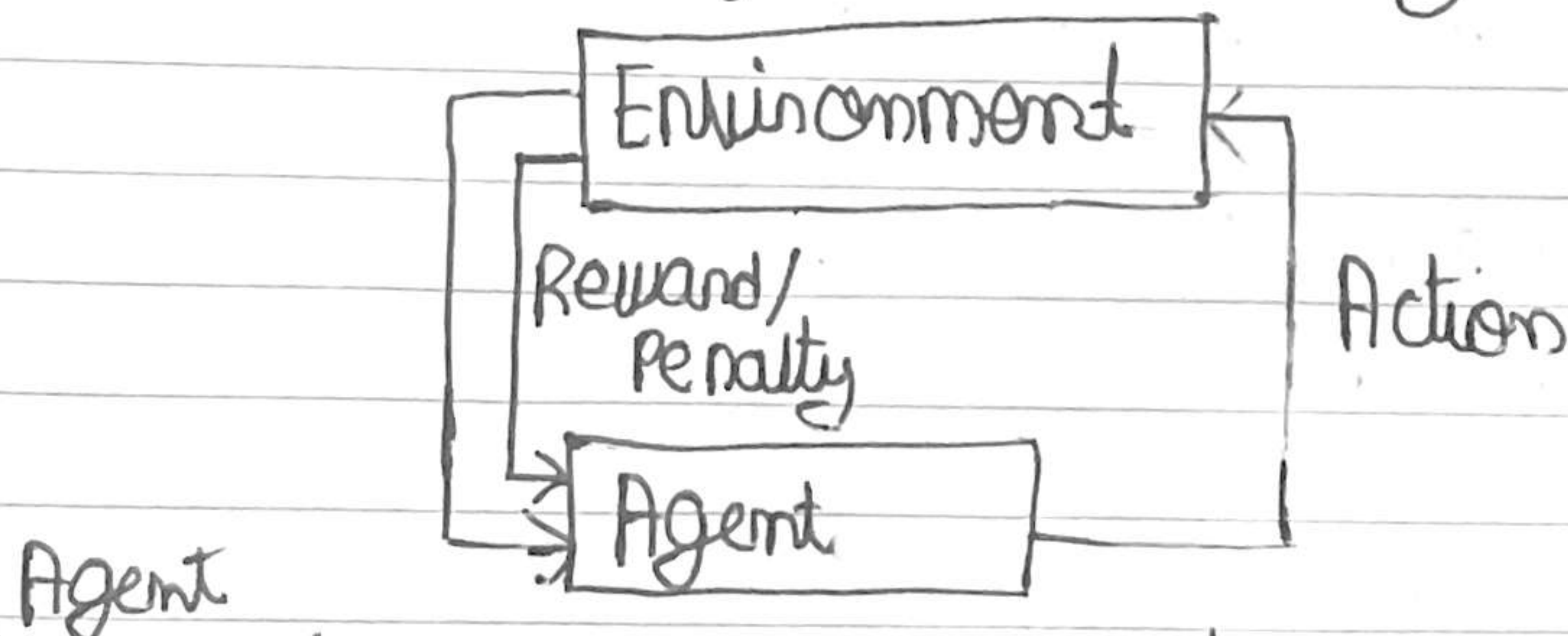
Model-Free RL-

- 1) No explicit model of the environment is created
- 2) Agent learns value functions or policies directly from interaction
- 3) Sub types:
 - Value based: Q -Learning, SARSA
 - Policy based: REINFORCE, Actor-Critic

Key Features of Reinforcement Learning-

- 1) Trial and Error learning: Agent explores environment to discover actions yield maximum results
- 2) Delayed Reward: Rewards may not be immediate. RL aims to maximize cumulative reward over time
- 3) Exploration vs Exploitation: Balance between trying new actions (exploration) and using known actions with high reward (exploitation)
- 4) Sequential Decision Making: Future states depend on current actions

Elements of Reinforcement Learning-



- 1) ~~Action~~ = Learner or decision maker
- 2) Environment = Everything the agent interacts with
- 3) State(s) = Representation of environment at a
- 4) Action (a) = choices available to the agent

- 5) Reward (π) = Strategy mapping states to actions
6) Value Function ($V(s), Q(s,a)$) = Expected cumulative reward from a state ~~of~~ or state action pair

Applications of Reinforcement Learning -

- 1) Game Playing / Video games
- 2) Robotics
- 3) Resource Management
- 4) Finance
- 5) Healthcare

Q. Exploration vs Exploitation trade off in reinforcement learning -

Ans - Exploration:

Trying new actions to discover their potential rewards

- Exploitation:

Choosing the action known to yield the highest reward based on past experience

- Tradeoff = Too much exploration $\rightarrow$ slow learning & lower immediate reward

Too much exploitation $\rightarrow$ risk of missing better long term rewards

- eg - A slot machine with multiple arms: the agent must try all arms (~~exploitation~~ exploration) before settling on the arm giving highest payout (exploit-
-ation)

Q. Calculate the Silhouette Score for a data point having $a=2.5$ (intra cluster distance) and $b=4.0$ (nearest cluster distance)

Ans Silhouette Score (S)

$$S = \frac{(b-a)}{\max(a,b)}$$

a = average intra cluster distance (within same cluster)

b = average nearest cluster distance (for the next closest cluster)

$$a=2.5$$

$$b=4.0$$

$$S = \frac{4-2.5}{\max(2.5, 4)} = \frac{1.5}{4} = 0.375$$

Moderate clustering quality

Business Value & Benefits of Association Rule Learning-

- 1) Customer understanding
- 2) Targeted Marketing
- 3) Improved Decision making
- 4) Inventory Optimization

FP Growth Algorithm-

1) Core concepts:

- Itemset: a group of one or more items analyzed together
- Support: Frequency of an itemset in transactions
- Frequent Itemset: Itemsets with support $\geq$ minimum threshold (min sup)

2) FP Tree Structure:

- Root node = starting point for all transaction paths
- Path nodes: represents individual items in each path
- Node count: shows how many transactions pass through each node

- Header Table: Links all occurrences of each item for quick access
- Provides high data compression

3) Step 1: Building the FP Tree

- First Pass: Scan the database once to count frequencies of individual items & create a sorted F-list
- Second Pass: Build the FP-tree by mapping transactions into paths. Shared prefixes compress data efficiently

4) Step 2: Mining Frequent Patterns

- Start from F-list Bottom: Process least frequent items first
- Conditional Pattern Base: Collect prefix paths for each item
- Conditional FP Tree: Build smaller FP trees recursively
- Recursive: Continue ~~with~~ until only single paths or Mining no frequent items remain.